

RESEARCH ARTICLE

Exceeding expectations increases satisfaction among Louisiana waterfowl hunters

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Abstract

Louisiana surveys waterfowl hunters every 5 years to assess hunter effort, harvest success, demographics, and preferences for proposed regulations and management practices. The 2020 Survey of Louisiana Waterfowl Hunters addressed hunter satisfaction patterns, hunter perceptions on waterfowl migration, associations of field experiences to satisfaction, associations among distribution waves, the efficacy of sample coverage by an email survey, and allowed comparison with hunter satisfaction with surveys in 2005, 2010, and 2015. We distributed a 31-question survey by email to 73,554 Louisiana waterfowl hunters following the 2019–2020 season. We received 13,483 total responses, of which 8,218 were usable (licensed respondents actively hunting waterfowl during the 2019–2020 season), resulting in a qualified response rate of 12.0%. We compared qualified respondents to the overall population of Harvest Information Program registrants (169,891) and found no significant differences for age class, geographic distribution, and license type. Our results indicated that surveys should be distributed in 3 waves, as the number of days hunted would be overestimated by 7% (mean of 15 days [95% confidence interval = 14.7–15.7] compared to a mean of 14 days [95% confidence interval = 13.7–14.4]) by a single survey. Hunters' satisfaction was independent of which geographic zone they hunted, whether they were members of

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conservation organizations, or how many waterfowl were seen. Our results also indicated that the probability of greater hunter satisfaction was increased by 98% as the number of waterfowl harvested surpassed 57, by 68% for respondents reporting that their actual harvest exceeded expected harvest by at least one bird, and by 61% for respondents reporting that their actual days hunted exceeded expected days hunted by at least one day.

KEYWORDS

email survey, hunter expectations, hunter satisfaction, Louisiana waterfowl hunters, survey methodology

Wildlife managers actively seek input from stakeholders to enhance management, encourage engagement, and comply with statute and legal tradition (Sax 1970, Riley et al. 2002, Smith 2011, Decker et al. 2014, Schummer et al. 2020). Meetings, hearings, and advisory boards can be used to collect public opinions, but a lack of tractable data and potentially biased input make public meetings a poor way to measure the opinions of stakeholders (Association of Fish and Wildlife Agencies and The Wildlife Management Institute 2019). Rather, scientifically-designed random surveys are recommended for obtaining stakeholder opinions (Riley et al. 2002, Vaske 2008, Dillman et al. 2009). Because of declining response rates to mail surveys, many public opinion surveys are now internet-based, which are quicker to implement, easier to analyze, and less expensive (Schonlau and Couper 2017, Campbell et al. 2018, Sullivan et al. 2023). Daikeler et al. (2020) and Sullivan et al. (2023) reported that email surveys can address objectives related to hunter effort and success and are an efficient and inexpensive alternative to random mail surveys.

Evaluation of satisfaction is key for the recruitment, retention and reactivation (R3) of waterfowl hunters (Schummer et al. 2020), and increasing participation in waterfowl hunting and wetlands conservation is an explicit objective of the North American Waterfowl Management Plan (2018). Retention of hunters can depend on satisfaction, and the current atmosphere of waterfowl hunting is commonly tumultuous (Whitaker et al. 2019, Iles 2020, Moorman 2020, Reynolds 2020, Wallen et al. 2020). North American breeding populations of waterfowl have been increasing (U.S. Fish and Wildlife Service 2023), but the proportion overwintering in states like Louisiana has been decreasing (Whitaker et al. 2019), potentially because of factors such as coastal wetland loss, sea level rise, and changing agricultural practices (Whitaker et al. 2019, Iles 2020, Moorman 2020, Reynolds 2020, Wallen et al. 2020). Cox et al. (2023) and Verheijen et al. (2024) recently confirmed northern shifts in average latitudinal migration, resulting in fewer ducks reaching the Gulf of Mexico coast. Moreover, while waterfowl hunter numbers are decreasing nationwide, many states including Louisiana are experiencing increased hunting pressure, including substantial participation by nonresident hunters (Iles 2020, Moorman 2020, Reynolds 2020, Wallen et al. 2020). Considering the challenges faced by Louisiana waterfowl hunters, it is of heightened importance to evaluate their satisfaction with regulatory frameworks, waterfowl management, harvest, and changes in migration. It is widely accepted that hunter satisfaction is driven by meeting and exceeding expectations (Brunke and Hunt 2007, Qazi et al. 2017, Schroeder et al. 2017, Bradshaw et al. 2019, Gruntorad et al. 2020).

The 2020 Louisiana Waterfowl Hunter Survey (LAWHS) was designed to address multiple objectives. Management objectives included the following: 1) updating waterfowl hunter participation, effort, attitudes, satisfaction, and demographics; 2) identifying regulatory preferences for proposed changes to existing hunting zones (Figure 1) and season splits; and 3) gathering opinions on Louisiana-specific regulatory practices (e.g., a new hunt option for military members, prohibiting spinning-wing decoys, and changing bag limits and season timing for some



FIGURE 1 Louisiana, USA waterfowl hunting zones for the 2019–2020 waterfowl hunting season. Regular season dates for ducks, coots, and mergansers included 60 days with one split: November 9–December 8, and December 21–January 19 in the Coastal Zone; November 16–December 8 and December 21–January 26 in the West Zone; and November 23–December 8 and December 14–January 26 in the East Zone. Two additional days were allocated for youth-only waterfowl hunting in each zone.

species). The 2020 LAWHS sought to explore the following: 1) hunters' perceptions on waterfowl migration and its relationship with satisfaction; 2) associations of field experiences to hunter satisfaction; and 3) associations of responses among distribution waves. Thus, relative to these objectives, and following the multifaceted-discrepancy model (Manfredo et al. 1995), we addressed 15 hypotheses organized into 3 constructs (migration, satisfaction, and sample coverage) in the 2020 LAWHS (Table 1). Here, we deployed a survey instrument to measure hunter satisfaction and its relationship to hunter attributes and address other human dimensions goals supporting conservation and management decisions in Louisiana.

METHODS

We obtained a list of 2019–2020 Louisiana Harvest Information Program (HIP) registrants from the U.S. Fish and Wildlife Service Division of Migratory Bird Management. Our sampling frame included all 2019 Louisiana resident and nonresident hunters who registered with HIP and had email addresses in their Louisiana Department of Wildlife and Fisheries (LDWF) license profiles. The email addresses included hunters who intended to hunt waterfowl, and hunters who intended to hunt dove (*Zenaida macroura*), snipe (*Gallinago delicata*), woodcock (*Scolopax minor*), or other migratory birds and may or may not have hunted ducks or geese.

TABLE 1 Fifteen research hypotheses tested in the 2020 survey of Louisiana (LA), USA waterfowl hunters. For each hypothesis, the alternative is no change or association.

Shortened title	Full statement of hypothesis
Migration construct	
1. Migration has decreased (by waterfowl zone)	1. Hunters perceive that waterfowl migration has changed negatively in LA (by waterfowl zone).
2. Migration has decreased (by NGO membership)	2. Hunters perceive that waterfowl migration has changed negatively in LA (by conservation organization membership).
Satisfaction Construct	
3. Satisfaction has decreased 2005–2020	3. Overall satisfaction of LA waterfowl hunters has decreased since 2005.
4. Satisfaction associated to expected migration	4. Overall satisfaction of LA waterfowl hunters is positively associated with expected numbers of migrating waterfowl.
5. Satisfaction associated to actual days hunted	5. Number of days hunted is closely associated to positive hunter satisfaction.
6. Satisfaction associated to waterfowl seen	6. Number of waterfowl seen is closely associated with hunter positive satisfaction.
7. Satisfaction associated to shots taken	7. Number of shots taken is closely associated with hunter positive satisfaction.
8. Satisfaction associated to actual harvest	8. Actual harvest is closely associated with hunter satisfaction.
9. Satisfaction associated to expected days hunted (days hunted index)	9. Overall satisfaction of LA waterfowl hunters is positively associated to the index of actual days hunted to expected days hunted.
10. Satisfaction associated to expected harvest (waterfowl harvested index)	10. Overall satisfaction of LA waterfowl hunters is positively associated to the index of actual waterfowl harvested to expected waterfowl harvested.
11. NGO members more satisfied	11. Members of conservation NGOs are more satisfied than nonmembers.
12. Hunters on private land more satisfied	12. Waterfowl hunters on private land are more satisfied than waterfowl hunters on public land.
Distribution wave construct	
13. Satisfaction decreases by wave of response	13. Overall hunter satisfaction decreases by wave of response.
14. Days hunted decreases by wave of response	14. Satisfaction with days hunted decreases by wave of response.
15. Harvest decreases by wave of response	15. Satisfaction with harvest decreases by wave of response.

We conducted the 2020 Louisiana Waterfowl Hunter Survey through an email distribution by the LSU Public Policy Research Lab following the 2019–2020 waterfowl season. The survey included 31 questions addressing hunting effort, methods, harvest, satisfaction, regulatory alternatives, self-identity, and demographics. We designed the survey using Qualtrics survey software (Provo, UT, USA), and vetted our survey through professional review by faculty and LDWF biologists prior to distribution. We delivered the survey in 3 waves post-season during the spring of 2020 (Dillman et al. 2009). An announcement email containing a link to the survey instrument was sent to 73,554 potential respondents on April 9, followed by a reminder email on April 16, and a last call email on April 23, 2020.

The waves were spaced at one-week intervals to maximize response rates and were limited to one response per Internet Protocol address (Dillman et al. 2009).

Response rate was calculated after removing respondents <16 years old, individuals who did not hunt waterfowl during the 2019–2020 waterfowl season, or individuals who did not answer ≥ 6 questions (i.e., respondents in these categories were deemed unqualified, therefore, a standardized cooperation response rate included only complete and partial responses and noncontacts [American Association for Public Opinion Research Standard Definitions; <https://www.aapor.org/Standards-and-Ethics>]). We generated 2 additional variables from response data to address hunter satisfaction: a days-hunted index (actual days hunted/expected days hunted), and a waterfowl harvested index (actual waterfowl harvested/expected waterfowl harvested). Index values >1.0 indicated respondents reported hunting more days than initially expected or harvesting more waterfowl than initially expected. Converting the data into indices was intended to more accurately capture the relationship of expectations and harvest with hunter's satisfaction (Vaske 2019). For the indices we removed outliers (identified as responses >3 standard deviations from the mean) from the days hunted index and waterfowl harvested index (Pukelshiem 1994, Lehmann 2013, Anhøj and Wentzel-Larsen 2018). We evaluated overall satisfaction with the hunting season using a 5-point scale, but to improve model fit (Agresti 2007) we also collapsed this into a 2-level binary satisfaction, which grouped respondents as either satisfied or dissatisfied ($n = 5,864$), and eliminated neutral responses ($n = 2,354$). For hypothesis testing, we used generalized linear models with appropriate link functions and response distributions as determined by $\hat{c}$ closest to 1, to identify the significance of individual variables. For all satisfaction hypotheses, we first analyzed binary satisfaction and also evaluated satisfaction using the 5-point scale. We set the significance threshold at $\alpha = 0.05$ following Dillman et al. (2009), Vaske (2011), Vaske et al. (2011), and Vaske (2019). For evaluating satisfaction across multiple survey years (Hypothesis 3; Table 1), we included the results of a survey of Louisiana waterfowl hunters conducted in 2005 (Miller et al. 2005), 2010, and 2015.

Dillman et al. (2009) noted that sample surveys require a statistical assessment for coverage error among the respondents within the sampled frame. To statistically assess coverage error, we compared all qualified responses for age class (16–25, 26–35, 36–45, 46–55, 56–65, and over 65 years old), geographic distribution (zip code region [9 in-state and 1 out-of-state], and license type [11], to the actual distributions of data in the overall population of HIP registrants [169,891]), using a multinomial distribution and a cumulative logit link function or binomial distribution and logit link, depending on whether the response was on a 5-point or 2-point (binary satisfaction) scale (PROC GLIMMIX, SAS 9.4, SAS Institute Inc, Cary, NC, USA). Additionally, we assessed the potential for hunter satisfaction to predict participation (days hunted) in an exploratory follow up analysis. Response variables were number of days hunted in state, total number of days hunted, index of days hunted, number of waterfowl bagged, and index of waterfowl bagged. Fixed effects were satisfaction with days hunted, satisfaction with waterfowl seen, satisfaction with shots taken, satisfaction with waterfowl harvested, and overall satisfaction with the season. For these analyses, we fit generalized linear models with a negative binomial distribution and a log link function (PROC GLIMMIX, SAS 9.4, SAS Institute Inc, Cary, NC, USA) that used the mean number of responses for each variable or interaction rounded to the nearest integer.

RESULTS

The 2020 LAWHS was sent to all HIP registrants in Louisiana with an email address in their LDWF HIP certification (73,554 out of 169,891 total registrants, or 45%), of which 68,578 were successfully delivered. We received 13,483 finished surveys yielding an overall response rate of 19.7%. We excluded 3,900 responses that did not hunt waterfowl last season, 2 under-age responses (age <16), and 1,383 responses that answered <6 questions to yield a qualified response rate of 12.0%. Qualified responses included 6,216 (75.6%) resident and 1,079 (13.1%) non-resident hunters, of which 7,200 (97.2%) were male hunters, and 190 (2.3%) were female hunters (percentages do not sum to 100.0% considering nonresponse for some questions).

Louisiana hunters, classified by primary geographic waterfowl zone hunted (Figure 1), perceived that numbers of waterfowl migrating to Louisiana have declined, reducing hunter satisfaction (Hypothesis 1). In the east zone, the baseline answer of numbers of migrating waterfowl have greatly decreased was more likely to be selected than migrating numbers have greatly increased (log odds -0.79 [± 0.46 SE], effect size [Cohen's d] = 0.88). In the coastal zone, similarly, the baseline answer of numbers of migrating waterfowl have greatly decreased was more likely to be selected than migrating numbers of waterfowl have greatly increased (log odds -0.76 [SE = 0.46], effect size [Cohen's d] = 0.88). In the west zone, the baseline answer of numbers of migrating waterfowl have greatly decreased also was more likely to be selected more than numbers of migrating waterfowl have greatly increased (log odds -0.55 [SE = 0.47], effect size [Cohen's d] = 0.80, $X^2_4 = 10.2$, $P = 0.04$ for all zones; Table 2). Similar effect sizes suggest that perceptions of decreased numbers of migrating waterfowl were similar in east and coastal zones and less prominent in the west zone. Louisiana hunters, classified by conservation nongovernmental organization membership (NGO member/nonmember), also perceived that numbers of waterfowl migrating have decreased in Louisiana (Hypothesis 2). Nongovernmental organization members were less likely to report greatly increased numbers of migrating waterfowl compared to greatly decreased migrating waterfowl than non-NGO members (log odds -0.13 [SE = 0.05], effect size [Cohen's d] = 0.74, $X^2_1 = 6.3$, $P = 0.01$; Table 2). Larger effect sizes for perceptions regarding changing numbers of migrating waterfowl by zone, rather than membership, suggest that geographic zone was more important than NGO membership.

On scale of 1 (very dissatisfied) to 5 (very satisfied), overall satisfaction of Louisiana waterfowl hunters has varied over time relative to previous 5-year surveys (Hypothesis 3). Satisfaction was low in 2005 ($\bar{x} = 2.01$), higher in 2010 ($\bar{x} = 3.57$), higher in 2015 ($\bar{x} = 3.46$), and lower in 2020 ($\bar{x} = 2.22$). Across survey years, hunters more frequently selected very dissatisfied compared to dissatisfied (log odds = 7.39 [SE = 1.51], effect size [Cohen's d] = 1.10), and dissatisfied compared with neutral or either level of satisfied (log odds = 3.24 [SE = 1.15], effect size [Cohen's d] = 1.10, $F_{4,5} = 15.64$, $P = 0.01$; Table 2). In the 2020 survey, overall satisfaction of Louisiana waterfowl hunters decreased with increased perception that numbers of migrating waterfowl have decreased (Hypothesis 4). More hunters selected very dissatisfied compared with dissatisfied, neutral, or both levels of satisfied (log odds = -2.59 [SE = 0.39] as perceived numbers of migrating waterfowl decreased, effect size [Cohen's d] = 1.20, $X^2_4 = 2019.17$, $P < 0.01$; Table 2).

Overall binary satisfaction decreased with total number of waterfowl seen with dissatisfied selected more commonly than satisfied (satisfied log odds = -0.03 [SE = 0.01] compared with dissatisfied, $t_1 = -2.35$, $P = 0.02$; Table 2). Satisfied was selected less frequently than dissatisfied as total number of days hunted increased (logit link, binomial model; log odds = -0.17 [SE = 0.01], $t = -9.21$, $P < 0.01$; Table 2). Lastly, overall binary satisfaction decreased with number of waterfowl harvested (logit link, binomial model) with satisfied selected less than dissatisfied (log odds = -0.10 [SE = 0.02], $t = -5.09$, $P < 0.01$). Our results indicated that 65.8% of avid hunters, in terms of above average days hunted and waterfowl harvested, remained dissatisfied despite apparent success in harvest, indicating that a more detailed examination of satisfaction across all 5 levels was warranted.

Overall satisfaction regarding the waterfowl season on the 1–5 scale was related to total days hunted (Figure 2B; Hypothesis 5), waterfowl harvested (Figure 2A; Hypothesis 8), total days hunted index (Hypothesis 9), and waterfowl harvested index (Hypothesis 10). Very avid hunters (>400 waterfowl harvested and >60 days hunted) more frequently (62%) expressed overall satisfaction of the waterfowl seasons as very dissatisfied or dissatisfied, and fewer (38%) of the very avid hunters indicated that their overall satisfaction was very satisfied or satisfied (Figure 2A,B). A greater number of days hunted increased hunter satisfaction with the overall waterfowl season on the 5-point scale (Hypothesis 5). Hunters who hunted more days more frequently selected very satisfied compared with satisfied (log odds 2.29 [SE = 0.29], effect size [Cohen's d] = 1.20) and somewhat satisfied compared with neutral or any level of dissatisfied (log odds 1.77 [SE = 0.20], effect size [Cohen's d] = 1.15, $X^2_4 = 135.06$, $P < 0.01$; Figure 2B; Table 2). A greater number of waterfowl seen was not associated with hunter satisfaction (Hypothesis 6, $P = 0.06$). A greater number of shots increases hunter satisfaction (Hypothesis 7) with hunters more frequently selecting satisfied compared with dissatisfied (log odds 4.67 [SE = 0.26], effect size [Cohen's d] = 1.32, $X^2_4 = 113.96$, $P < 0.01$; Table 2). Increased harvest increased hunter satisfaction (Hypothesis 8). Hunters were more likely to select very satisfied compared with the other 4 answers as harvest increased (log odds

TABLE 2 Results of generalized linear models testing 15 research hypotheses addressing migration perception, satisfaction (5-point scale and binary), and responses across 3 distribution waves ($n = 8,218$) in the 2020 survey of Louisiana, USA waterfowl hunters sponsored by the Louisiana Department of Wildlife and Fisheries. Binary satisfaction collapsed the 5-point scale to satisfied or dissatisfied. For models, the baseline for estimating log odds was very dissatisfied (5-point scale) or dissatisfied (binary).

Hypothesis (Model type)	<i>n</i>	Log odds (SE)	Goodness of fit $\hat{C}$	Type III <i>P</i>	Conclusion
1. Migration has decreased, by waterfowl zone (Cumulative logit link, multinomial model).	6,558	East -0.79 (± 0.46 SE) Coastal -0.76 (± 0.46 SE) West -0.55 (± 0.47 SE)	1.117	0.04	1. Hunters agree that waterfowl migration has decreased, irrespective of zones hunted.
2. Migration has decreased, by NGO membership (Cumulative logit link, multinomial model).	6,558	-0.13 (± 0.05 SE)	1.117	0.01	2. Hunters agree that waterfowl migration has decreased, irrespective of conservation organization membership.
3. Satisfaction has decreased 2005–2020 (Logit link, beta distribution).	20	Very dissatisfied 7.39 (± 1.51 SE) Somewhat dissatisfied 3.24 (± 1.15 SE)	3.97	0.01	3. Dissatisfaction was highest and statistically significant in 2005 and 2020.
4. Binary satisfaction associated with expected migration (Cumulative logit link, multinomial distribution).	6,420	-2.59 (± 0.39 SE)	2.128	<0.01	4. Hunters who believe that migration has decreased are dissatisfied.
5. Satisfaction associated to actual days hunted (Cumulative logit link, multinomial distribution).	5,098	Very satisfied 2.29 (± 0.29 SE) Somewhat satisfied 1.77 (± 0.20 SE)	1.070	<0.01	5. Satisfaction increases as the number of days hunted increases.
6. Binary satisfaction associated with waterfowl seen (Cumulative logit link, multinomial distribution).	5,098	1.57 (± 0.28 SE)	1.070	0.06	6. No significant differences.
7. Binary satisfaction associated with shots taken (Logit link, binomial distribution).	5,098	4.67 (± 0.26 SE)	1.070	<0.01	7. Satisfaction increases as the number of shots taken increases.
8. Binary satisfaction associated with actual harvest (Cumulative logit link, multinomial distribution).	5,098	3.99 (± 0.31 SE)	1.070	<0.01	8. Satisfaction increases as the number of waterfowl harvested increases.

(Continues)

TABLE 2 (Continued)

Hypothesis (Model type)	n	Log odds (SE)	Goodness of fit $\hat{C}$	Type III P	Conclusion
9. Binary satisfaction associated with expected days hunted (days hunted index; logit link, binomial distribution).	4,322	0.44 (± 0.06 SE)	1.315	<0.01	9. Satisfaction increases as the index of actual to expected days hunted increases.
10. Binary satisfaction associated with expected harvest (waterfowl harvested index; logit link, binomial distribution).	4,931	0.79 (± 0.02 SE)	1.433	<0.01	10. Satisfaction increases as the index of actual to expected waterfowl harvest increases.
11. Conservation NGO's more satisfied (binary satisfaction; logit link, binomial distribution).	5,098	0.06 (± 0.12 SE)	1.070	0.60	11. No significant difference in satisfaction based on conservation organization membership.
12. Hunters on private land more satisfied (binary satisfaction; logit link binomial distribution).	1,778	0.01 (± 0.01 SE)	1.253	0.35	12. No significant differences.
13. Satisfaction decreases by wave of response (binary satisfaction; logit link, binomial distribution).	5,864	0.14 (± 0.08 SE)	NA	0.08	13. No significant differences.
14. Days hunted decreases by wave of response (logit link, binomial distribution).	5,596	-0.10 (± 0.01 SE)	1.101	<0.01	14. Number of total days hunted decreases by wave of response.
15. Harvest decreases by wave of response (logit link, binomial distribution).	6,144	0.00 (± 0.01 SE)	1.263	0.79	15. Number of total waterfowl harvested does not decrease by wave of response.

3.99 [SE = 0.31], effect size [Cohen's d] = 1.30, X^2_4 = 135.06, P < 0.01; Table 2). Overall satisfaction of LA waterfowl hunters was positively associated with waterfowl harvest index >1.0, indicating the respondents harvested more days than planned or expected and harvesting more birds than expected increased satisfaction (Hypothesis 9). Hunters were satisfied more frequently than dissatisfied as the waterfowl harvest index increased >1.0 (log odds 0.79 [SE = 0.02], effect size [Cohen's d] = 0.96, X^2_1 = 29.97, P < 0.01; Table 2). Overall binary satisfaction of Louisiana waterfowl hunters was positively associated with days hunted index >1.0 indicating the respondents hunted more days than planned or expected (Hypothesis 10). Satisfied was selected more frequently than dissatisfied as the days hunted index increased >1.0 (log odds 0.44 [SE = 0.06], effect size [Cohen's d] = 0.84, X^2_1 = 51.46, P < 0.01; Table 2). Members of conservation NGOs had no difference in satisfaction compared to nonmembers (Hypothesis 11; P = 0.60). Waterfowl hunters on private land had no differences in satisfaction than hunters on public land (Hypothesis 12; P = 0.35).

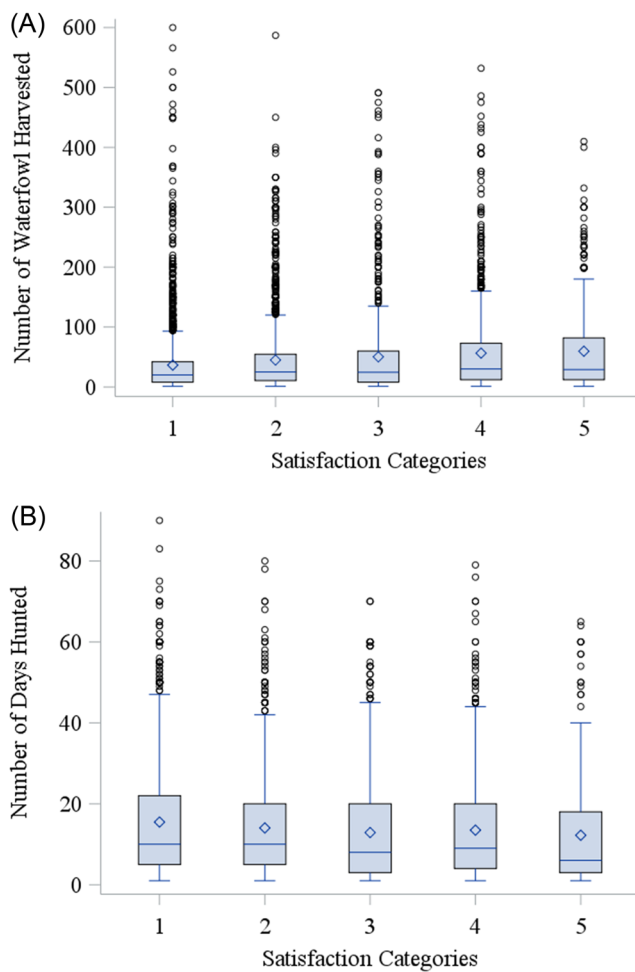


FIGURE 2 Overall satisfaction with the 2019–2020 Louisiana waterfowl season (1 = very dissatisfied, 2 = dissatisfied, 3 = neither dissatisfied nor satisfied, 4 = satisfied, 5 = very satisfied) compared to reports of total number of waterfowl harvested (A) and total number of days hunted (B) for waterfowl during the 2019–2020 Louisiana, USA waterfowl hunting season. Bars indicate 95% confidence intervals. Many avid hunters (>400 waterfowl harvested and >60 days hunted) expressed low satisfaction (62%), and no (0) waterfowl hunters reporting over 450 waterfowl harvested (A) or over 75 days hunted (B) responded as very satisfied. Counts may include Louisiana early teal season (15 days), regular waterfowl season (60 days), and conservation order for light geese (38 days).

Across 3 waves of responses, overall binary satisfaction did not decrease (Hypothesis 13) with hunters selecting satisfied slightly more than dissatisfied across waves (log odds 0.14 [SE = 0.26], effect size [Cohen's d] = 1.09, $X^2_4 = 5.06$ $P = 0.08$), and binary satisfaction with waterfowl harvested did not decrease (Hypothesis 15) with no differences in selecting satisfied compared with dissatisfied across waves (log odds 0.00 [SE = 0.01], effect size [Cohen's d] = 0.04, $X^2_1 = 2.18$, $P = 0.79$; Table 2). However, binary satisfaction with number of days hunted decreased by wave of response (Hypothesis 14). Hunters selecting satisfied compared with dissatisfied decreased per wave (log odds -0.10 [SE = 0.01], effect size [Cohen's d] = 0.74, $X^2_4 = 18.31$, $P < 0.01$; Table 2).

We did not detect any evidence of coverage bias as there were no differences between survey respondents and the overall HIP population in age class ($F_{1,6} = 0.04$, $P = 0.85$), geographic distribution by zip code region

($F_{1,18} = 1.87$, $P = 0.19$), or license type ($F_{1,12} = 0.98$, $P = 0.34$). We observed that the survey data underrepresented age classes 16–25 (survey 7.7% compared with HIP 16.7%) and >65 (survey 11.0% compared with HIP 16.9%). The survey overrepresented age classes 26–35 (survey 20.8% compared to HIP 16.2%), 36–45 (survey 21.2% compared to HIP 14.8%), and 46–55 (survey 21.2% compared to HIP 14.8%).

DISCUSSION

More than 80% of Louisiana waterfowl hunters were concerned that the number of waterfowl migrating to Louisiana is decreasing, irrespective of which waterfowl zone they primarily hunted or whether they were members of conservation organizations. The concerns were compelling and with merit, as published reports continually indicate that fewer waterfowl are migrating to Louisiana (Whitaker et al. 2019, Iles 2020, Moorman 2020, Reynolds 2020, Wallen et al. 2020). Cox et al. (2023) and Verheijen et al. (2024) used banding and harvest data to independently confirm northern shifts in average winter ranges, resulting in fewer mallards (*Anas platyrhynchos*) and other ducks reaching Louisiana. A perception of fewer waterfowl in Louisiana may in part explain why dissatisfaction was high in the 2020 survey. Compared to historical hunter surveys in Louisiana, the widespread dissatisfaction reported by Miller et al. (2005) was linked to changes in farming practices (69% of respondents) and warmer winters (61% of respondents), which might imply concerns about changes in waterfowl distributions; however, that survey did not directly address migration. The specific perception of decreased numbers of waterfowl migrating to Louisiana was widely discussed online, at public hearings, and in private meetings, motivating a special publication by the Louisiana Department of Wildlife and Fisheries (LDWF 2020). It may be that hunters were voicing their frustration with a less-than-expected return (harvest) on time and money invested. Nevertheless, voicing dissatisfaction appears to be a persistent and pervasive cultural phenomenon in the Louisiana waterfowl hunter experience, as we observed some meaningful proportion of respondents reporting dissatisfaction irrespective of survey year (80% in 2005, 45% in 2010, 22% in 2015, and 65% in 2020), including many hunters with high numbers of days in the field and/or high rates of harvest success (Miller et al. 2005, Laborde 2014, Laborde et al. 2015, Sullivan 2021). Potentially, this consistent dissatisfaction may be related to unmeasured factors (e.g., lease costs, gasoline costs, crowding). However, over the multiple survey efforts many potential sources of dissatisfaction have been included (e.g., farming practices, season and split timing, and access), and have not identified the cause of dissatisfaction experienced by hunters, suggesting that this expression may be related to deeply held expectations, based on historical experiences, both personal and storied, within the Louisiana waterfowl hunting community. This reinforces that expectations are a component of satisfaction and warrants additional research into the nature of these expectations. Indeed, we found that hunter satisfaction increased when expectations were met and exceeded. As Bradshaw et al. (2019) noted, unrealistic expectations can yield poor satisfaction, and potentially unrealistic expectations among Louisiana waterfowl hunters (e.g., based on high numbers of breeding waterfowl and historical migration patterns), resulting in hunters who are consistently disappointed and express dissatisfaction. Therefore, like Brunke and Hunt (2007) and Bradshaw et al. (2019), our results suggest that managers should ask hunters about expectations and assist hunters with setting appropriate expectations. Managers should help hunters understand the challenges, and the opportunities provided by seasonally changing climatic conditions and long-term changes in habitat conditions through increased agency outreach using social and conventional media.

Success in field performance (days hunted, shots taken, and waterfowl harvested) and meeting and exceeding expectations (indices of actual to expected days hunted and actual to expected harvest) increased satisfaction, consistent with reports from similar studies (Brunke and Hunt 2007, Qazi et al. 2017, Schroeder et al. 2017, Bradshaw et al. 2019, Gruntorad et al. 2020). However, unlike findings reported in other studies, hunters' satisfaction was not associated or was weakly associated with how many waterfowl they saw, where they hunted (public vs. private land), or whether they were members of any conservation organizations (Brunke and Hunt 2007, Qazi et al. 2017, Schroeder et al. 2017, Bradshaw et al. 2019, Schroeder et al. 2019). Gruntorad et al. (2020)

reported that seeing game is an important attribute related to hunters' satisfaction across hunting 4 different game species, which is consistent with similar studies (Schroeder et al. 2017, Bradshaw et al. 2019, Schroeder et al. 2019). Despite the consistent substantial relationship between seeing game and satisfaction throughout published literature, we found that hunters' perceptions of migration numbers outweigh their satisfaction with the actual number of waterfowl they observed.

Human dimensions research and management decisions are premised on the assumption that sample surveys are representative of the target population (Battaglia et al. 2016, Haziza and Beaumont 2017). Our results support the findings of Sullivan et al. (2023) in that a widely distributed email survey can produce respondents that are representative and not statistically different from the overall population of HIP registrants in the population of Louisiana hunters. Our survey verified the representativeness of all 3 demographic variables tested (age class, zip region, and license type). Like Schmidt-Catran et al. (2019), our results indicated that satisfaction did not vary across waves; however, the total days hunted decreased by wave of response, indicating that more avid hunters responded more quickly, which also parallels previous research (Laborde 2014, Bradshaw et al. 2019). Our finding reinforces the need for multiple distribution waves in survey design if it is important to study objectives to accurately reflect hunter effort.

Based on the findings from the exploratory analysis of reported satisfaction (5-point scale) to effort and success, many Louisiana waterfowl hunters who reported higher satisfaction (very satisfied and satisfied) with the number of waterfowl harvested (direct measurement and harvest index), and with the number of days hunted (direct measurement and days hunted index), were those hunters who actually harvested more waterfowl and hunted more days. Therefore, for the average hunter, more days hunted and more waterfowl harvested was a strong indicator of higher satisfaction, especially when exceeding some *a priori* target or expectation. However, our most avid hunters, at least by measures of many waterfowl harvested, high waterfowl harvested index, many days hunted, and high days hunted index, were very dissatisfied. Therefore, hunting effort and harvest poorly explained satisfaction for the most avid hunters, which is further supported by our analysis of binary satisfaction where increased harvest and days hunted decreased satisfaction. Our findings also suggest that hunters could be satisfied with the number of waterfowl they have seen but may become dissatisfied if they are not able to harvest them. The simple association that successful waterfowl hunters expect to harvest more waterfowl could suggest a factor contributing to the consistent pattern of some very successful hunters expressing dissatisfaction when their expectations are not fulfilled, a recurrent theme across surveys and years. Our findings reinforce the need for managing both expectations and harvest opportunity, as more satisfied hunters generally hunt more frequently and we hypothesize will hunt more consistently across years.

Our study demonstrates that satisfaction might be more effectively quantified through indices, which relate actual to expected experiences. Our results identified an inverse relationship between harvest and overall hunter satisfaction when measured in binary terms of satisfied or dissatisfied. Quantifying the factors influencing overall satisfaction remains a challenge with potential unquantified explanations found in perceptions of migration, missed harvest opportunities, and/or just the norms of finding something to complain about within the hunting community. Therefore, further research into the components of overall satisfaction is needed. Future research could also expand the investigation of indices by pairing data from pre- and post-season surveys for analysis.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ETHICS STATEMENT

Survey protocols ensured informed consent, anonymity, and confidentiality of responses, and were approved by the LSU AgCenter Institutional Review Board (IRB Protocol No. HE-19-05).

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SUPPORTING INFORMATION

Additional supporting material may be found in the online version of this article at the publisher's website.

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